FINITENESS AND FINITENESS OF THE QUOTIENT BY THE INTEGRAL CLOSURE FOR A REDUCED ONE-DIMENSIONAL NAGATA LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Finite length of a module supported at a single point. Stacks Project, Algebra, Lemma algebra-lemma-support-point (tag 00L5). Turns the support condition on A'/A into finiteness of its length.
- The element of the maximal ideal whose localization makes A and its normalization agree. Stacks Project, Varieties, Lemma lemma-pre-pre-delta-invariant (tag 09NK). Supplies the element f whose existence drives the support computation.

Source reference. Original reference 03GH: sources/stacks-chow/tags.txt, source lines 4528–4528. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 030C: sources/stacks-chow/tags.txt, source lines 3943–3943. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00FR: sources/stacks-chow/tags.txt, source lines 581–581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 032T: sources/stacks-chow/tags.txt, source lines 4032–4032. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3R: sources/stacks-chow/tags.txt, source lines 15201–15201. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00L5: sources/stacks-chow/tags.txt, source lines 775–775. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a reduced Nagata 1-dimensional local ring. Let A' be the integral closure of A in the total ring of fractions of A. Then A' is a normal Nagata ring, $A \to A'$ is finite, and A'/A has finite length as an A-module.*

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The total ring of fractions is essentially of finite type over A hence $A \to A'$ is finite because A is Nagata, see Algebra, Lemma 03GH. The ring A' is normal for example by Algebra, Lemma 030C and 00FR. The ring A' is Nagata for example by Algebra, Lemma 032T. Choose $f \in \mathfrak{m}$ as in Lemma 0C3R. As $A' \subset A_f$ it is clear that $A_f = A'_f$. Hence the support of the finite A-module A'/A is contained in $\{\mathfrak{m}\}$. It follows that it has finite length by Algebra, Lemma 00L5. $\square$